

Semester Project: Domain Discovery, Recommendation, and Graph Intelligence

Milestone: Recommendation, Ranking, and Predictive Decision Engine (Week 10)

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1. Task Framing

Task type: Recommendation.

Given a user's positive interaction history (games endorsed via `voted_up=True`), the system predicts which unseen game they would also endorse. Output: ranked list.

Clustering results from Week 7 (6 segments: Casual, Action, Local Multiplayer, Narrative/JRPG, VR, Strategy) feed into this pipeline as candidate-filtering features.

2. Data Alignment

All systems operate on the same 500-game universe from `data/processed/steam_v1.parquet`. AppID is the common join key:

- `steam_v1.parquet` - user records (`author_steamid`, `AppID`, `voted_up`, `playtime`)
- `games_may2024_full.csv` - game metadata (`name`, `tags`, `genres`, `categories`)
- `artifacts/clustering/cluster_labels_k6.npy` - cluster labels from Week 7

3. Evaluation Protocol

- Interaction matrix: sparse binary, users x games. `voted_up=True` -> 1.
- Qualifying users: ≥ 2 distinct positive interactions.
- Train/test split: one positive per user held out (`random_state=42`).
- Primary metric: HR@10 - fraction of users whose held-out game is in top-10.
- Leakage prevention: training items masked to `-inf` before ranking.

Statistic	Value
Positive interactions	442,652
Unique users	426,601
Unique games (universe)	484
Qualifying users (eval)	12,389
Matrix density	0.0078%

4. Baseline 1 - Popularity

Design: All games ranked by aggregate 'recommendations' count. Same ranking for every user.

Justification: Non-personalized ceiling. Any personalized system failing to beat it is unjustified.

Limitation: Biases toward AAA titles; cannot surface niche Indie games.

5. Baseline 2 - Content-Based

TF-IDF on `name` + `description` + `tags` + `genres` + `categories` (1-2 grams, 3,000 features).

TruncatedSVD -> 50D. User vector = centroid of training game vectors. Ranking by cosine similarity.

Limitation: Overspecializes to known preferences; ignores collective behavioral signal.

6. System 3 - Matrix Factorization (SGD)

Objective:

$$\min_{\{P,Q\}} \sum_{(u,i)} (R_{ui} - p_u^T q_i)^2 + \lambda(\|P\|^2 + \|Q\|^2)$$

SGD updates per pair (u,i):

$$p_u \leftarrow p_u + \eta(e_{ui}q_i - \lambda p_u)$$

$$q_i \leftarrow q_i + \eta(e_{ui}p_u - \lambda q_i)$$

$$\text{where } e_{ui} = R_{ui} - p_u^T q_i$$

Parameter	Value	Justification
Learning rate eta	0.03	Standard for implicit binary data
Regularization lambda	0.01	Prevents overfitting on 99.8% sparse matrix
Epochs	30	Sufficient for convergence in 500-game universe
Latent dim k	{10,20,50}	Swept via HR@10; best k=10 used for final model
Init	N(0,0.01)	Small random values to break symmetry

7. System 4 - Hybrid Recommender

$$s_{\text{hybrid}}(u,g) = \alpha * s_{\text{content}}(u,g) + (1-\alpha) * s_{\text{MF}}(u,g)$$

Both scores min-max normalized to [0,1]. alpha in {0.1, 0.3, 0.5, 0.7, 0.9}.

Hybrid is legitimate: AppID is the common key for both content matrix and MF item factors. Both systems index games identically - no cross-catalog alignment issue.

8. Offline Evaluation Results

Generated by python src/recommendation.py with random_state=42.

System	HR@5	HR@10	HR@20
Popularity baseline	0.4597	0.5796	0.7124
Content-based baseline	0.1806	0.2170	0.2669
MF k=10	0.1103	0.1888	0.2949
MF k=20	0.0764	0.1333	0.2183
MF k=50	0.0756	0.1235	0.1998
Hybrid (alpha=0.3, k=10) BEST	0.2057	0.2569	0.3330

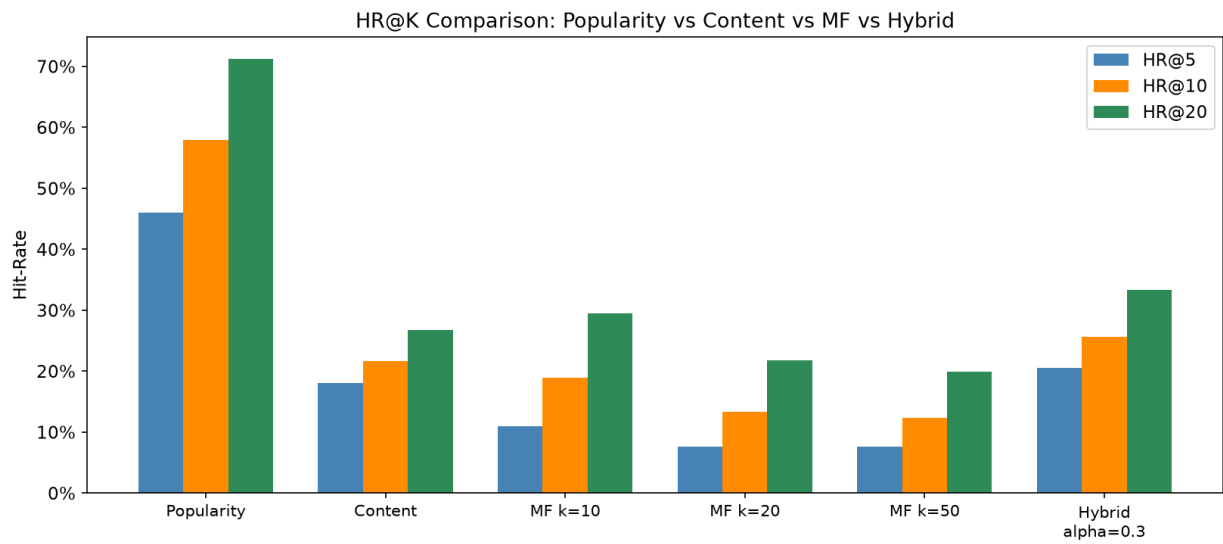


Figure: HR@K comparison. Hybrid ($\alpha=0.3$) achieves best personalized HR@10=25.7%.

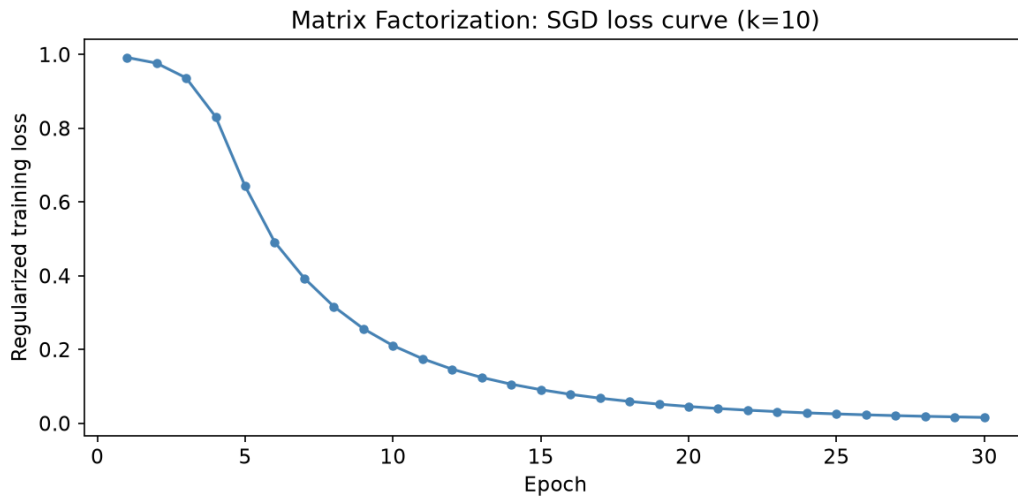


Figure: SGD loss per epoch (k=10). Monotonic decrease confirms convergence within 30 epochs.

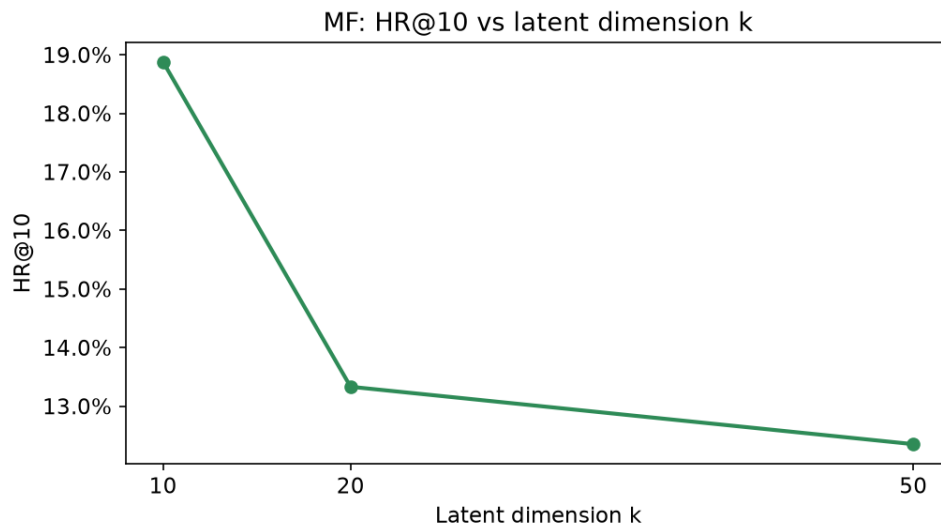


Figure: HR@10 vs latent dimension k. k=10 generalizes best on this sparse 484-game universe.

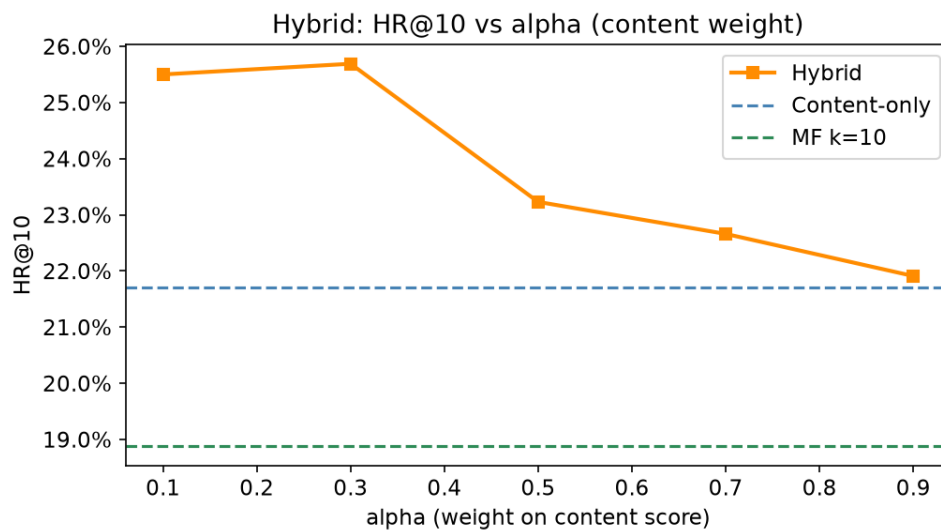


Figure: Hybrid HR@10 vs alpha. Optimal alpha=0.3 (30% content + 70% MF).

9. Error Analysis

Strong cases (hits): Users with history concentrated in one cluster get accurate recs. MF identifies similar narrow-profile users.

Failure cases (misses): Cross-cluster taste - history spans clusters (e.g., Action + Narrative) but held-out game belongs to a third cluster. Latent factors average out competing signals.

Secondary failure: Popularity contamination in users with only 2 qualifying interactions.

10. Reproducible Pipeline

```
python src/ingest.py          # Step 1: data ingestion
python src/feature_engineering.py # Step 2: feature matrices
python src/clustering_week7.py  # Step 3: clustering (Week 7)
python src/recommendation.py    # Step 4: recommendation engine
python src/generate_figures.py  # Step 5: report figures
```

All outputs deterministic with random_state=42.

11. Ethics and Access Note

- CC0 license. No PII beyond public Steam IDs used as opaque row indices.
- author_steamid: opaque identifier. No real names, emails, or financial data.
- Recommendation fairness: popularity bias present. Future: inverse propensity scoring.